

The Price of Thought

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Takeaways

- The cost of AI performance has been falling 30–65% per quarter in the last 2–3 years, with different rates for different benchmarks. That is equivalent to 77–98%/year. These figures assume one is using the cheapest model up to the job at any given time,
- Extrapolating to the Epoch Capabilities Index, an overall indicator of LLM strength, the price of artificial thought has fallen 61%/quarter. That compounds to almost 98%/year: a nearly fifty-fold cost drop every twelve months.
- Those numbers are averages. The pace of decline appears to vary by benchmark, year (even month), and the level of performance one is costing. One common pattern: on many benchmarks, the cost falls fastest near the state-of-the-art performance level. When a new model sets an all-time record on a benchmark, it wins a premium price in the marketplace. But competitors soon approach its performance, and prices plunge.
- If this pace continues for many years—a big “if”—it will far exceed price drops seen for other transformative technologies. The price of thought is currently falling 5.6 times faster than did the price of compute in the 1980s and 1990s, 26 times faster than the price of lithium batteries since 1991, and 100 times faster than the price of lighting since 1800.

The repo is [here](#). GitHub pages are built from it; go [here](#) to explore many plots and tables.

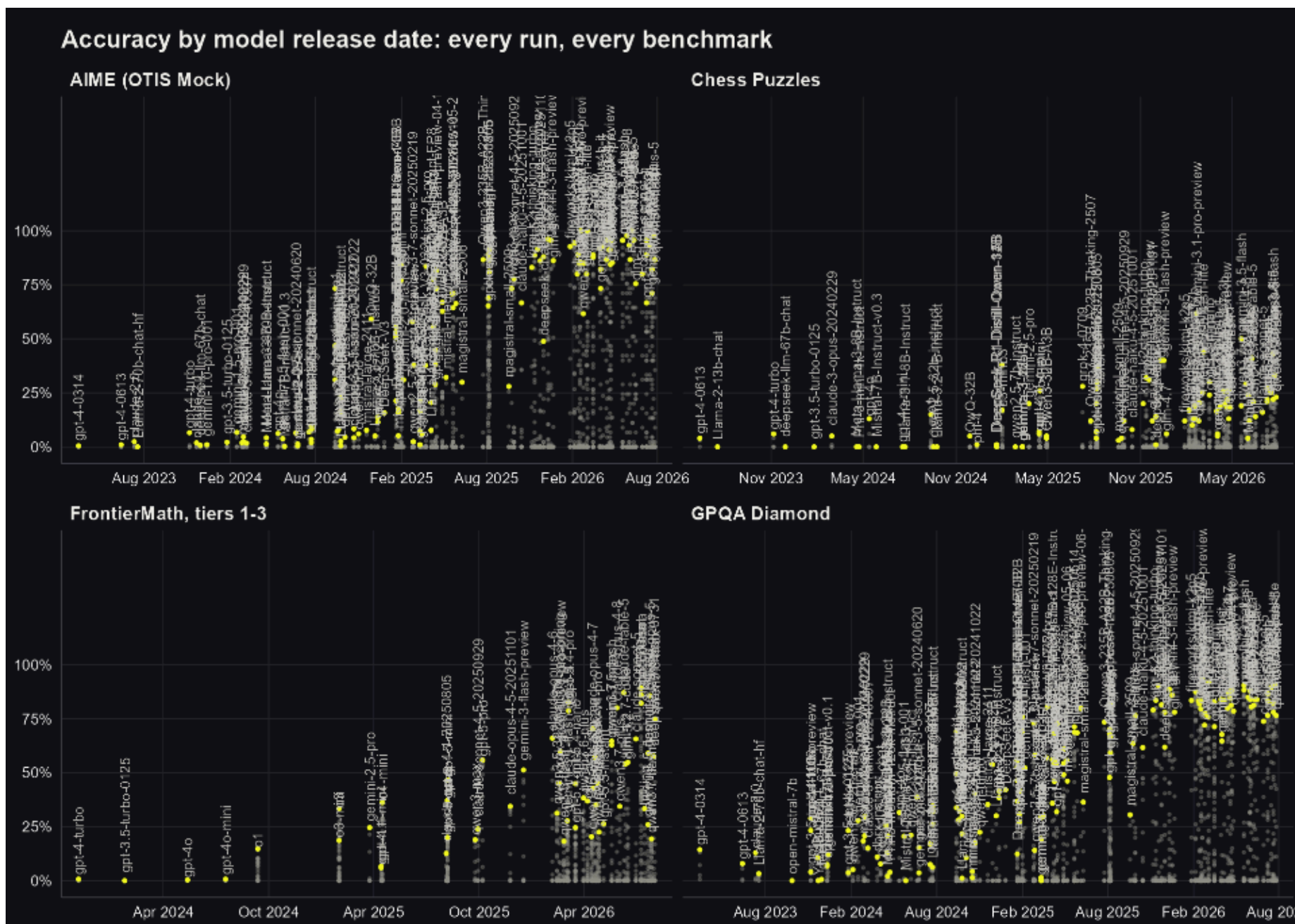
Data

Luke and JS described the data and frame the modeling problem [here](#).

At my request, Luke added an option into the code that produces the “truncated” results, to change how unanswered questions are scored. Instead of being coded as 0, they are coded with the probability of guessing the correct answer. So, for example, in the GPQA Diamond benchmark, whose questions are four-way multiple-choice, if a model only answers one question before running out of budget, the rest are scored as if (in expectation) the model guessed on the rest. Then, before modeling, GPQA scores are clipped to 0.25 and rescaled from [0.25, 1] to [0, 1].

Figure 1 on the next page gives a rough sense of the data for four benchmarks of primary interest, plotting measured performance of a model versus its date of release. Of the four benchmarks, Chess Puzzles remains the most difficult.

Figure 1. Accuracy versus release date for four benchmarks



Modeling

The challenge

For each benchmark, we have a set of triples (a_i, t_i, c_i) representing the accuracy of a model's answers (between 0 and 1), the model's release date, and the dollar cost of the run.

We want to characterize the rates of decline in the price of a given level of performance—not of the average model, but of the best available at any given time and price point.

To be concrete, we define two empirical Pareto frontiers, for accuracy and cost. The empirical accuracy frontier is the highest performance achieved within a given budget using models available up to a given date:

$$A_t(c) = \max_{i|t_i \leq t, c_i \leq c} a_i$$

The empirical cost frontier is the lowest cost at which at least one available model can meet or beat a given accuracy target, again restricting to models released by a given date:

$$C_t(a) = \min_{i|t_i \leq t, a_i \geq a} c_i$$

The accuracy frontier can only rise with time and the cost frontier only fall. (More precisely, each is pointwise weakly monotonic in the indicated direction.)

The statistical modeling problem is challenging in a few ways:

1. Output is bounded above, since accuracy can't exceed 100%. In most economic models of production and cost frontiers, output can go arbitrarily high.
2. The frontier is constructed as a maximum or minimum at each point. This makes it unstable. A single benchmark result—or a single missing one—can substantially shift the frontier for months. Most of the time we use statistics to study averages or other measures of central tendency. When doing so, each additional data point affects results less than the last—on average—and the estimate of the average converges. When studying extrema, this limiting behavior may either not set in, or set in more slowly. So we should presume that the empirical frontiers are noisy representations of “true” frontiers.
3. Because of #2, the sampling program matters more. E.g., if a higher percentage of newer models are benchmarked, this will tend to puff out the more regions of the frontiers. Because testing models takes time and money, I understand that we have

mostly tested models that appear to be near the frontier as of their release date. The sampling is therefore a bit ad hoc. That's not a criticism. It just means that we don't have a credible, clean model of the data generating process, in a context where it might substantially affect the results.

4. Within the time frame studied, change is extremely fast. Most studies of production frontiers take place in less dynamic industries, like manufacturing and farming, where one can imagine a large number of producers tracing out the frontier at any given time. Here,

Items 2 and 3 together mean that when we model the frontier, it's hard to produce credible standard errors, thus estimates of statistical significance. For those are always founded on a model of the data generating process, which we think or hope is reasonable, including in its specification of noise components.

The models

I wound up fitting five main models to the data—two just to the empirical frontiers and three to all the available data. I actually did the latter first, but two of those models are harder to explain and I no longer prefer their results, so I'll swap the order of explanation.

It seems most natural to model performance on a benchmark as a function of spending and time: spending is an input to production, time affects the technology of production, and performance is the output. But our ultimate interest here is in determinants of cost, not performance. And there is a long tradition in economics of viewing production and cost functions as duals, and modeling both with similar methods.

Ideally our answers should not depend on whether we model cost of a given level of performance or performance for a given cost, since either way we are estimating the same relationships among variables. And once we have fit a model to one frontier we can back out the implications for the other (see below). In practice, there are some asymmetries (or anti-symmetries) between how the same modeling approach is applied to the two frontiers. Consider: in fitting a line to a cloud of points in the plane, one could minimize the vertical distances between the points and the line or the horizontal distances, and thus get somewhat different answers. My intuition is that when models are nonlinear, as most here are, results from the two perspectives can diverge more in practice.

The approach here was to do both and use a web interface to make the results easily browsable.

Model 1: Pareto logistic/least squares

The idea here is to fit smooth surfaces to the empirical frontiers defined above. Like lines through a cloud of dots, the best fits will have real data points on both sides.

For the *accuracy* model, we assume that at any given time, frontier model accuracy is related to both log cost and release date according to an S curve:.

$$E(a|x_a) = \sigma(x_a' \beta_a)$$

x_a is a vector of controls—at a minimum, log cost ($\ln c$), release date (t), and a constant. “ $E(a|x_a)$ ” means the expected/average accuracy of a model given values for those controls. β_a is a vector of coefficients. σ is the logistic S curve function, which rises from 0 toward $-\infty$ to 1 toward $+\infty$. The parameters are estimated with a “fractional logit” regression (Papke and Wooldridge 1996), whose maximand has the same mathematical formula as the logit model for binary outcomes, and is reinterpreted to allow any outcome values between 0 and 1, inclusive.

For the *cost* model, we model log cost, which is unbounded, as linear in its predictors:

$$E(\ln c|x_c) = x_c' \beta_c$$

x_c contains, at a minimum, the logit/inverse logistic of a , symbolized $\sigma^{-1}(a)$, along with t and a constant. This is fit with ordinary least squares. Because the logit transform maps 0 and 1 to $\pm\infty$, unlike in the previous model, scores of 0 or 1 are dropped from the sample. That includes cases when a model scores at or below the expected score from guessing. For example, if an LLM scores 0.25 on GPQA Diamond, the rescaling of the score range from $[0.25, 1]$ to $[0, 1]$ will give that model a 0.

These models are fit once, across two-dimensional data spaces: the dimensions are time and cost for the accuracy model, and time and accuracy for the cost model. In the fits, the frontier surface is represented not just by the record-setting data points that define its corners, but by 100×100 grids. The grid points are evenly spaced values of release date, log cost, or accuracy. The use of a grid reflects the idea that the absence of a new record at any given time—the absence of a data point—is itself information. Still, if standard errors were reported for these regressions, then this gridding could be problematic since arbitrarily raising the resolution of the grids would increase the sample size, and thus, naively, reduce standard errors.¹

¹ This problem might be handled with methods for spatially autocorrelated data.

Because entire Pareto *surfaces* are being modeled, any particular slice of the best-fit surface, such as for accuracy versus cost as of release date 2026-01-01, will generally *not* be the best fit just for the corresponding slice of the empirical frontier. The quality of fit in any slice is constrained by the goal of finding the best overall fit across all slices, subject to the parameterization.

Model 2: Envelope

This model is the same as the previous one except that

- It forces the theoretical frontier to lie completely outside the empirical frontier—above when modeling accuracy and below when modeling cost.
- In the accuracy model, some benchmark tests produce perfect scores of 1. (And, as explained, 0's and 1's must be dropped only in the cost models, not the accuracy models.) But it is impossible for a smooth S curve envelope to ever reach 1, thus for the envelope to embrace such a point of perfection. So before fitting, 1's are replaced with $1 - 1/2n$ where n is the number of questions we challenged the model to answer (number of runs times number of questions per run, column "n_samples" in the "cost_truncated_curves.csv" data file). In other words, perfect scores are replaced with a number that represents getting all questions right half the time and all but one right the other half.

The requirement that the envelopes stay outside all of the corners of the empirical frontier staircases (or just touching them) constitutes a large set of nonlinear constraints on the parameters. Sequential Least Squares Programming (SLSQP) is used to find the constrained best fits.

Model 3: Full-data logistic/least squares

These are like the Pareto logistic/least squares except they are fit to all the test results in the data set, rather than a grid of points representing the staircase-like empirical frontiers. While they are not structurally focused on the frontier, they are included for a few reasons. They are not as intrinsically unstable since they are not based on the min and max operators. They are simple. And a priori they could give about the same answers as models of the frontiers. I would expect that if we modeled the history of RAM prices in \$/megabyte over the last 50 years, the rate of decline would be about the same whether or not we explicitly modeled the frontier. But here of course, the time span is much shorter and the progress faster.

Model 4: Stochastic frontier analysis

In economics, one established approach to modeling production and cost frontiers is stochastic frontier analysis (SFA). While in our case there is only one input to production—money—usually SFA models have multiple inputs, like capital and labor, whose supply is jointly bound by a budget constraint. For a given budget, the production frontier is the outer boundary on the domain of possible output levels across different mixes of, say, capital and labor. Most farms and firms will lie inside the boundary, because of inefficiency relative to the technological limit.

In SFA statistical models, output is a function of input(s), plus *two* error terms. One is conventional and symmetric: it captures random factors in the output variable, such as the output of firms just being high or low by chance, or being mismeasured as such. The other error term is distinctive and asymmetric: it represents the inefficiency of individual producers. Since it enters production functions with a negative sign (or cost functions with a positive sign), it is itself always non-negative. One firm, for example, could have an inefficiency of 10% and another 20%.

An SFA production model looks like

$$y = x'\beta + v_i - u_i$$

The symmetric error term, v_i , is assumed to be normally distributed around 0 with variance σ_v^2 . The asymmetric inefficiency term, u_i , can take one of several forms; here it is assumed to be independently distributed as the positive half of a normal distribution with variance σ_u^2 .

The mental model for the SFA production specification is that there is some frontier representing the limits of the possible, and while some firms lie outside it because of flukes in measurement or what happened at the farm in a particular month, many more are on the interior side. And because they are striving for efficiency as one approaches the frontier on the interior side, the density of firms rises. This picture implies that there is a clear but sharply truncated ridge in the data, where firms bunch up against the frontier. If it's strongly present, then model fitting can detect and separate the two error terms, i.e., the two causes of deviation from the theoretical frontier.

Within the SFA tradition, our modeling context is unusual in that output—performance on a benchmark—is bounded above. So I had Claude estimate a version of SFA that looks like

$$y_i = \sigma(\theta_i + v_i) \text{ with } \theta_i = x_i'\beta - u_i$$

Claude realized that expressing this as an objective function (log likelihood) to be maximized would require integrating out u_i . If each firm's inefficiency, u_i , were known rather than uncertain and random, then the u_i could be held constant during model fitting and this would be a standard logistic model. It would be fit to the data by finding β to maximize the sum of the log likelihoods of all observations. That is, it would maximize

$$\ln \mathcal{L}|u: = \sum_i [y_i \ln \theta_i + (1 - y_i) \ln(1 - \theta_i)]$$

where $\ln \mathcal{L}|u$ means the log likelihood when taking all the u_i as given.² Since the u_i are in fact assumed to be independently distributed according to the half-normal distribution with variance σ_u^2 , the log likelihood *unconditional* on them is the expected value of the above over the possible values of the u_i :

$$\ln \mathcal{L} = \sum_i \ln \int_0^\infty du_i f(u_i; \sigma_u^2) \exp[y_i \ln \theta_i + (1 - y_i) \ln(1 - \theta_i)]$$

where $f(\cdot; \sigma_u^2)$ is the probability density for u_i under the half normal.

In fact, that objective is not quite the one maximized. In an attempt to stabilize the model fits, u_i is assumed to be the same for each *run*, by which I mean a set of data points all synthesized from a full run by simulating various budget caps. So, indexing runs by r and individual observations by r, i , the precise log likelihood is

$$\ln \mathcal{L} = \sum_r \ln \int_0^\infty du_r f(u_r) \exp \sum_i [y_{ri} \ln \theta_{ri} + (1 - y_{ri}) \ln(1 - \theta_{ri})]$$

Claude chose to estimate the integrals using (non-adaptive) quadrature, which is a popular numerical technique. Also, to improve precision and speed of the iterative search for the optimal parameters, I had Claude write code to compute, at each trial set of parameter values, the gradient of $\ln \mathcal{L}$ with respect to all of the parameters.

² Technically, I have moved here from assuming v_i as normal to assuming it has the logistic cumulative distribution function. I think the normal-based model, the fractional probit, makes more sense theoretically since by the Central Limit Theorem, normal distributions naturally arise as mixtures of partially uncorrelated random processes. But the logistic distribution is widely used in our context, e.g., in Item Response Theory, and in practice the distinction rarely matters.

Perhaps not surprisingly, this cleverness ultimately did not produce convincing results. While the model fitting might theoretically be able to detect a moving frontier, in practice it seems that the frontiers are moving too fast, and the distribution of benchmark tests in their wake too sparse, for the model to reliably distinguish the two error components. For example, for the chess puzzles benchmark, the model finds almost no progress over time, which looks like a mathematically degenerate result.

Model 5: Stochastic frontier analysis with time-dependent inefficiency spread

The final model is a variant on the main SFA model. It starts from the visualization that as the production frontier races across time-cost space, the distribution of *available* models, which includes all older models, is increasingly dispersed behind it. So the model allows σ_u^2 to depend on release date. In particular $\ln \sigma_u$ is linear in a constant and t .

This change does not improve matters. Now, for example, the fit says that the frontier on the AIME benchmark has hardly moved.

Nonlinear models

The base models are linear in log cost and release date. For the accuracy models, this means assuming that the passage of each quarter, and each doubling of spending, causes the same improvement in the logit of benchmark accuracy. For cost models, it assumes that the passage of each quarter, and each one-unit increase in the logit of performance, has the same proportional impact on cost.

These assumptions are not obviously true. Cost declines might be accelerating or decelerating in calendar time. There might be diminishing returns to spending at any given time.

To allow for nonlinearities, all models are also run in quadratic variants. In the accuracy models, the regressor vector x gains t^2 , $(\ln c)^2$, $t \times \ln c$. In the cost models, it gains t^2 , $(\sigma^{-1}(a))^2$, $t \times \sigma^{-1}(a)$.

While mathematically natural, a peculiarity of quadratic models is that they can imply parabolic trends, with directional changes, like models getting worse over time at certain times or cost levels. Ordinarily these implausible projections appear outside or on the fringes of data range. But it can still produce model fits that look strange and are hard to explain.

An alternative approach is apply the Box-Cox transform to each of the x variables. For example, for cost, we would apply

$$\phi_{\lambda}(c) = \frac{c^{\lambda}-1}{\lambda}$$

This introduces the exponent λ as an additional parameter to be estimated. The Box-Cox transform shifts and scales the power transform ($c \rightarrow c^{\lambda}$) in order to give it some nice properties. When $\lambda = 1$, it is just $\phi_{\lambda}(c) = c$, so it includes linear models as a special case. When $\lambda > 1$ it is convex and when $\lambda < 1$ it is concave. And in the $\lambda \rightarrow 0$ limit, it equals $\ln c$.

In the models with Box-Cox transforms, each of the predictor variables is allowed its own λ exponent. When brought to the accuracy variable, the transform is applied to the odds ratio $a/(1 - a)$. That way the transform embraces $\ln[a/(1 - a)]$ as a special case, which is the standard logistic function, as used in the linear model. In addition, just as in the quadratic models, the *product* of the two Box-Cox-transformed terms is also entered as a regressor. Here, the same λ exponents are carried over.

A complication in terminology: the “Box-Cox transform” was used by [Box and Cox \(1964\)](#) to transform dependent variables, to make their distributions more normal-like. Here we are actually following [Box and Tidwell \(1962\)](#) in applying it to the independent variables. So these regressions are more probably described as Box-Tidwell.

The benefit of the Box-Tidwell approach is that it allowed nonlinearity while still assuring that the relationship to each regressor is monotonic.

The fitting code conducts the search for the best fit with a two-level process. In the upper level, a search is conducted in the space of possible λ pairs. For each set of trial values, the other model parameters are estimated using the same methods as for the linear models while holding the λ exponents fixed.

Preferences among the models

Because of our interest in cost declines at the frontier, I now favor the models that are fit to the empirical frontier rather than the full data set. The SFA models fail to credibly identify a (moving) frontier, and the simple logistic regression on the full data set is not designed to find it.

Of the two models fit just to the frontier, I prefer the first, which is not constrained to lie outside the empirical frontier. Our interest is not in where some theoretical frontier lies, beyond the reach of most models at any given point in time, but in real-world cost-performance tradeoffs. Our data are quite representative of actually available performance because they cover essentially all the important models. A smooth approximation to the empirical frontier surface distills its angular contours into a few numbers.

As for the order of the model, I prefer Box-Tidwell. In the 3-D surface plots, the linear fits often look unrealistic. Quadratic models do better, but sometimes reverse themselves around the edges of the data. The Box-Tidwell fits follow the data more closely while looking more realistic because of their monotonicity.

Extracting estimates of cost declines

The linear cost models directly provide estimates of the impact of time (release date) on log cost, in the form of a coefficient β_t .

To extract the equivalent from linear accuracy models takes more work. The model fits give rise to isoquants, curves in the cost-time plane along which predicted accuracy is constant. Movement along these iso-accuracy curves is characterized by

$$da = \frac{\partial a}{\partial t} dt + \frac{\partial a}{\partial \ln c} d \ln c = 0$$

Solving for $d \ln c / dt$,

$$\frac{d \ln c}{dt} = - \frac{\frac{\partial a}{\partial t}}{\frac{\partial a}{\partial \ln c}} = - \frac{\beta_t}{\beta_{\ln c}}$$

where the β 's are coefficients from an accuracy model fit.

Extrapolating individual benchmark results to the ECI

The ECI [is built](#) on the equation

$$a_{m,b} = \sigma(\alpha_b(C_m - D_b))$$

$a_{m,b}$ is the accuracy of model m on benchmark b , C_m is the capability of model m , normalized so that Sonnet 3.5 gets a 130 and GPT-5 a 150. D_b is the difficulty of a benchmark, on the same scale. The “discrimination” parameter α_b is a benchmark-specific conversion rate from ECI capability units to logits of performance on the benchmark.

This model comes out of Item Response Theory, which is the dominant method for assessing the difficulty of standardized tests and scoring students who take them. Here, though individual benchmark results rather than individual questions are the unit of observation. The core idea is that how well a model performs on a benchmark depends only on the difference between its capability, C_m , and the benchmark's difficulty, D_b . The α_b

values determine how sensitive the expected score is to that difference, i.e., how quickly a series of ever-better models will climb the S-curve from poor to excellent performance.. If α_b is low, then it takes a lot of capability improvement to climb the curve, but there is a wide range of capability levels over which further improvement will noticeably improve scores. On the other hand, if α_b is high, then once models approach a certain capability threshold ($C_m \approx D_b$), they will cross rapidly from poor to excellent performance. Soon after, the benchmark will be saturated, and further increases in capability will be hard to detect with this benchmark.

Suppose a linear accuracy model for performance on the OTIS Mock AIME benchmark produces an estimate of β_t , which is how much accuracy rises per year, in logit terms. To convert that impact to ECI units, we divide by the discrimination parameter α_b :

$$\frac{\beta_t}{\alpha_b}$$

where b is now taken to be the index for the AIME benchmark. (Recall that *multiplying* by α_b converts the other way, from ECI units to standard logit units.

Since we fit the same models separately for multiple benchmarks, we will in this way get multiple estimates of how the passage of time—or an increase in spending—affects frontier ECI performance. How should we combine these estimates?

A good way, Claude persuades me, is to average them, weighting by the α_b^{-2} . Why? It is generally natural when combining multiple estimates of the same thing to weight them by their *precision* (inverse variance). For example, as [Dario explained in 2011](#), when updating a normal prior with normally distributed evidence, the posterior mean is the average of the means of the prior and the evidence, weighting by the precision of each. Here, α_b^{-2} roughly proxies for precision in the sense that when α_b is small, a modest bit of noise in our estimate of benchmark performance—or in the effect of time or spending on performance—corresponds to a large uncertainty on the ECI scale. If $1/\alpha_b$ scales the noise then $1/\alpha_b^2$ scales the variance and α_b^2 scales its inverse, precision.

Thus, given multiple estimates, via multiple benchmarks, of the impact of time on ECI performance, we form the weighted average

$$\bar{\beta}_{t,ECI} = \frac{\sum_b \alpha_b^2 \frac{\beta_t}{\alpha_b}}{\sum_b \alpha_b^2} = \frac{\sum_b \alpha_b \beta_t}{\sum_b \alpha_b^2}$$

And likewise for the impact of $\ln c$.

Then, we can estimate the proportional rate of cost decline in frontier ECI performance with

$$- \frac{\bar{\beta}_{\ln c, ECI}}{\bar{\beta}_{t, ECI}}$$

That is done here, taking α_b 's from the “estimated_slope_scale” column of [edi_scores.csv](#).

Currently five, primary benchmarks are incorporated: AIME, Chess Puzzles, Frontier Math Tiers 1–3, GPQA Diamond, and Mystery Game Puzzles.

Results

First-order: average rates of decline

Most of the results are presented through the [GitHub Page](#). I'll focus here on estimates of overall rates of cost decline, by benchmark. Unsurprisingly, this will validate the model preferences I voiced above.

Table 1 shows many estimates of the average quarterly rate of cost decline. The first two columns actually show other things: the date at which data start for each benchmark, and the α_b^2 weights.

After those, the first result column shows the average quarterly drop in the cost of frontier performance estimated using *none* of the foregoing models. Here, instead, non-parametric estimates are computed by averaging empirical cost declines over a large set of points in data space. In particular, the code creates 100×100 grids in time and accuracy. In the accuracy dimension, the range is scaled to stay within the best accuracy yet achieved by any model at the given date. For each point, the minimum cost is computed for achieving accuracy at least as high from models released by that date. The computation is repeated for the time point one quarter later. The resulting cost ratios are then averaged geometrically and subtracted from 1.

Among the five primary benchmarks, the decline rates from this exercise range from 35%/quarter for Chess Puzzles to 51%/quarter for FrontierMath tiers 1–3.

The following columns show which models manage to produce realistic results. The models have the disadvantage of imposing a certain mathematical structure, which is never exactly correct. On the other hand, to the extent the models are correct, they will be more precise.³

In the second column, we see that fitting a linear model to the empirical log cost frontier—linear, that is, in t and $\sigma^{-1}(a)$ —produces *reasonable* results for most benchmarks. But the differences from the non-parametric numbers are nontrivial: 10–20 points, except for a near-perfect match on the slowest-changing Chess Puzzles.

The next column is unusual within the table because it turns to a nonlinear specification—the Box-Tidwell model for the cost frontier, the preference for which was motivated earlier. In this model the rate of cost decline depends on both the current cost and the release date, so no one parameter constitutes an overall bottom-line on the rate of cost decline. To distill the model’s overall implications, we average over the same grid of cost and accuracy values as before. This time what is averaged is the model’s predictions for the instantaneous rate of cost decline at each grid point. The matches to the non-parametric numbers is now much better; and while it is not obvious a priori which is more correct, the convergence between them tends to validate both.

Fits of the other models generally deviate much more from those just reviewed, in ways that look dubious. Requiring the predicted cost frontier to lie perfectly outside of (or just touching) the frontier—to envelope it—again produces some implausible values, like 2.7%/quarter for SWE-Bench Verified. Modeling *all* of the data, not just the points along the empirical cost frontier, produces whacky results, with many negative signs indicating cost increases. So does the base SFA model.

Turning to models for accuracy (last two columns), the logistic model for the frontier, which is linear in the sense of just depending on log cost and time, produces more stable values, albeit ones that are distinctly higher than in the first few columns. Requiring this fit too to envelope tends to lift the numbers a bit more.

Making some of these worse-performing models nonlinear surely helps them mold to the data surface and produce more reasonable results. But I prefer a model that works reasonably well even when restricted to linearity.

The bottom row of the first panel aggregates the results for the primary AI models and reexpresses them in ECI points. **The Box-Tidwell fit suggests that the price of ECI performance has fallen 61%/quarter.**

³ If a two-dimensional data set is indeed generated by the classical linear model—a straight line plus an independent error term—then estimating the line’s slope by fitting a model embodying that structure will be more precise than calculating a lot of slopes between various points and averaging them.

In communicating these results, Epoch might want to compare them to price drops for other technologies. Updating [Nordhaus \(1997\)](#), [Fouquet and Pearson \(2006\)](#) and [Fouquet \(2026\)](#), provide a data series on the price of a lighting—pure illumination—back to 1300. Since 1800, the price to produce a million lumen-hours has fallen 3.7%/year. Nordhaus (2007, Table 8) estimates that the price of computation fell 74.3%/year in the 1980s and 60.8%/year in the 1990s (logarithmic growth rates). Lithium batteries [dropped in price](#) from \$9,200/kilowatt-hour in 1991 to \$78 in 2024, or about 15%/year.

Thus, the price of thought, proxied by ECI performance, has fallen 100 times faster than lighting has, 26 times faster than that of lithium batteries, and 5.6 faster than computation did in the 1980s and 1990s.⁴ Of course, the track record for AI is only a few years, and this speed of decline may not continue.

Second-order: how the rate of decline varies

The [plot viewer](#) includes interactive 3-D surface plots of cost and instantaneous rates of cost decline, as well as heat maps that are equivalent to viewing the surfaces from above. The 3-D plots of the *cost* surface also include the empirical frontiers, so you can compare the empirical and modeled frontiers. (On the left, choose “Cost” for the modeled variable, “Cost” or “Cost decline rate” for the plotted variable, and “3-D” under Rendering.)

A glance at these plots makes clear that there is diversity across benchmarks in the relationship of cost to time and accuracy. Some of the apparent diversity might just be an artifact of fitting a flexible data to a sample that is too narrow in the time or accuracy dimension. But some of it is plausibly real.

Figure 2 is a screenshot capturing the cost decline heat maps for four benchmarks. Yellow indicates the fastest declines and purple the slowest. To give a sense of where the model is extrapolating out of the data range, staircases are overlaid on the plot to show the strongest and weakest performance yet realized by any model. For three of the benchmarks—AIME; FrontierMath tiers, 1–3; and GPQA Diamond—it appears that the cost declines fastest near state of the art (SOTA) performance. One scenario that would explain this pattern: when a new, SOTA-defining model appears, its maker can briefly charge a premium for this performance. But competitors quickly catch up, and the price of such performance plunges. Eventually this dynamic fades and the decline slows, even in percentage terms.

⁴ All computations carried out in log points. The 61.3%/quarter drop for the ECI equates to $-4 \times \ln(1 - 0.613) \times 100 = 562$ log points/year. The price of light [fell](#) an average 3.7 log points/year, from £8497 in 1800 to £2.15 in 2023 (after inflation adjustment). The two Nordhaus figures average to 67.6 log points. The fall for batteries is $\ln(78 / 9210) / (2024 - 1991) = 14.4$ log points/year.

This pattern can be quantified by returning to the models that generate the plots. The Box-Tidwell model for the cost frontier is

$$\ln c = \beta_a \phi_{\lambda_a} \left(\frac{a}{1-a} \right) + \beta_t \phi_{\lambda_t} (t) + \beta_{at} \phi_{\lambda_a} \left(\frac{a}{1-a} \right) \phi_{\lambda_t} (t) + \beta_1$$

Differentiating that with respect to t and then negating gives the rate of cost *decline*. We are interested in turn in how that varies with respect to accuracy level, expressed in logits, which is a second derivative. It works out to:

$$\frac{\partial}{\partial \sigma^{-1}(a)} \left(- \frac{\partial \ln c}{\partial t} \right) = \beta_{at} t^{\lambda_t - 1} \left(\frac{a}{1-a} \right)^{\lambda_a}$$

In nearly all the [Box-Tidwell cost model fits](#), $0 < \lambda_a < 1$. This tells us that according to the fits, at any given time, the rate of cost decline is a power law in the odds ratio $a/(1 - a)$ with a small exponent λ_a . Going from an accuracy level of 0.5 to 0.75, for example—from an odds ratio of $0.5/(1-0.5) = 1$ to $0.75/(1 - 0.75) = 3$, multiplies the rate of cost decline by 3^{λ_a} . Taking the AIME benchmark as an example, λ_a is estimated at 0.193. $3^{0.193} \approx 1.23$, suggesting that the price of 75% accuracy falls 23% faster than the price of 50% accuracy.

Table 1. Average quarterly rate of cost decline, holding accuracy fixed

Benchmark	Data start	ECI weight	Non-parametric	Cost models				Accuracy models		
				Model frontier, require envelopment		Model all data	Stochastic frontier analysis	Model frontier	Model frontier, require envelopment	
				Model frontier	Box-Tidwell				Linear	Linear
<i>Model order</i>				Linear	Box-Tidwell	Linear	Linear	Linear	Linear	Linear
<i>Primary models</i>										
AIME (OTIS Mock)	2023-03	0.0496	61.3%	43.4%	71.8%	28.5%	-1.2%	-9.7%	68.8%	75.8%
Chess Puzzles	2023-06	0.0143	35.4%	35.1%	33.1%	27.9%	3.9%	6.7%	57.1%	51.9%
FrontierMath, tiers 1-3	2023-11	0.0286	67.5%	51.1%	67.7%	56.4%	24.0%	26.8%	72.0%	71.3%
GPQA Diamond	2023-03	0.0143	58.3%	35.9%	59.6%	4.4%	-8.4%	-10.2%	61.2%	80.2%
Mystery Game Puzzles	2023-06	0.0282	50.6%	39.2%	40.3%	4.6%	5.0%	-8.9%	55.5%	48.5%
<i>Pooled (ECI-weighted)</i>			58.2%	42.7%	61.3%	29.4%	5.8%	1.1%	65.5%	70.6%
<i>Secondary models</i>										
FrontierMath 2025-02	2024-05	0.0670	62.2%	42.1%	46.7%	9.3%	-8.0%	-54.4%	56.9%	60.5%
FrontierMath, tier 4 (2025-07)	2024-06	0.0815	37.6%	28.1%	23.8%	-0.0%	-33.3%	-231.4%	64.9%	50.5%
FrontierMath, tier 4 v2	2025-01	0.0792	61.6%	54.3%	54.4%	54.4%	30.3%	26.6%	78.5%	84.6%
MATH, level 5	2023-06	0.0321	61.2%	49.3%	67.9%	15.0%	-12.0%	-22.1%	70.1%	81.7%
SimpleQA Verified	2024-10	0.0063	39.5%	40.1%	42.6%	14.2%	-5.3%	-30.2%	61.3%	99.0%
SWE-Bench Verified	2024-05	0.0051	30.4%	25.3%	31.1%	2.7%	1.5%	-3.1%	36.0%	40.3%

All numbers are estimates of the average quarterly drop in the cost of a given level of accuracy on a given benchmark, over the years of available data. The bottom row of the first panel aggregates the primary models using the weights shown, which are α_b^2 where the α_b are estimated parameters in the ECI IRT model.. The "non-parametric" values are averages over even 100×100 grids that span the benchmark's release date and accuracy ranges, with the latter scaled within the state of the art (SOTA) for that benchmark at each time point. At each point, the lowest cost of performance at least as good one quarter later is found and divided by the initial cost. The geometric mean of the ratios is taken and subtracted from 1. Results in all cost model columns are coefficients on release year as an explanator for log cost, again reexpressed quarterly; except that in the nonlinear Box-Tidwell variant, the statistic is computed as in the non-parametric column, using instantaneous decline rates at the grid points implied by the best fit. In the accuracy model columns accuracy is the dependent variable and log cost as an explanatory variable, so the rates are extracted as -b_t/b_x. "Model frontier" means modeling the empirical frontier as realized at a grid of points, either with ordinary least squares (OLS; for log cost) or with a logit link (for accuracy). "Model frontier, require envelopment" means the same, but with the constraint that the fitted surface is never above any data point (for cost) or below (for accuracy). "Model all data" means modeling all runs, not just the frontier, with OLS. "Stochastic frontier analysis" models the frontier and the distribution of runs around it with a half-normal distribution of inefficiency.

Figure 2. Heat maps of cost decline rate vs. release date and accuracy for four benchmarks, Box-Tidwell model of cost frontier

